ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

MECHATRONICS

PRACTICE EXAM 6

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 3. Laid out like the DHET Mechatronic Systems paper: FIVE questions of 20 marks. Sensors, relays and power supplies, actuators, PLCs and ladder logic, hydraulic power packs and a press circuit, electro-hydraulic timing circuits, and maintenance and fault-finding.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL calculations. Draw all circuits and symbols neatly, using a ruler and standard symbols.
5. Write neatly and legibly.

QUESTION 1: SENSORS AND ELECTRONICS

1.1 Name THREE types of sensors used in mechatronic systems, and state what EACH one detects. (6)

1.2 Explain how a relay allows a low-current control circuit to switch a high-current circuit. (4)

1.3 Draw a labelled block diagram of a regulated DC power supply. (5)

1.4 Explain the function of the capacitor in the filter section of a power supply. (2)

1.5 An oscilloscope shows a waveform with a period of 20 ms. Calculate its frequency. (3)

[20]

QUESTION 2: ACTUATORS AND PLCs

2.1 Define the term actuator and give TWO examples. (4)

2.2 Explain how the direction of rotation of a three-phase induction motor is reversed. (2)

2.3 Name FOUR main parts of a programmable logic controller (PLC). (4)

2.4 Draw a PLC ladder diagram for a conveyor motor with: a START button (I0.0, NO); a STOP button (I0.1, NC); an overload contact (I0.2, NC); the motor contactor (Q0.0) that latches; and a run lamp (Q0.1) that is on while the motor runs. (6)

2.5 Explain what is meant by the scan cycle of a PLC. (4)

[20]

QUESTION 3: HYDRAULICS

3.1 Define the following terms:

3.1.1 Viscosity (1)

3.1.2 Pressure (1)

3.1.3 Flow rate (1)

3.2 Name FOUR main parts of a hydraulic power pack and explain the function of EACH. (8)

3.3 Explain how a one-way flow control valve controls the speed of a cylinder. (3)

3.4 Design a hydraulic circuit for a workshop press. Make use of: a double-acting cylinder; a lever-operated 4/3-way valve with a closed centre; a one-way flow control valve that slows down the extending stroke (meter-out); a pressure relief valve; and a pressure gauge. (6)

[20]

QUESTION 4: ELECTRO-HYDRAULIC CONTROL

4.1 Explain the purpose of a limit switch in an electro-hydraulic circuit. (2)

4.2 Explain what an on-delay timer relay does. (3)

4.3 A press cylinder is controlled by a 4/2-way double-solenoid valve (Y1 extends, Y2 retracts). Design the electrical control circuit so that: pressing START (S1) makes the cylinder extend; when the cylinder is fully extended, limit switch S2 starts an on-delay timer T1 that holds the press closed for 5 seconds; after the 5 seconds, the cylinder retracts; and an emergency stop removes the power from the control circuit. (11)

4.4 Give FOUR advantages of electro-hydraulic control over purely manual hydraulic control. (4)

[20]

QUESTION 5: MAINTENANCE AND FAULT-FINDING

5.1 Differentiate between preventive maintenance and breakdown maintenance. (4)

5.2 Explain, step by step, how to replace a leaking hydraulic hose safely. (7)

5.3 A solenoid valve does not switch over when the START button is pressed. Give FOUR possible causes. (4)

5.4 Give FIVE ways in which dirt and other contamination can get into hydraulic oil. (5)

[20]

TOTAL: 100

END OF QUESTION PAPER